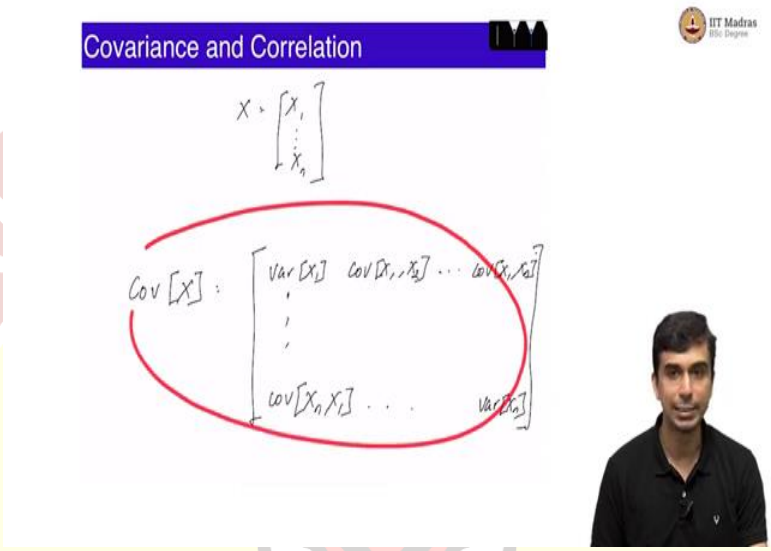


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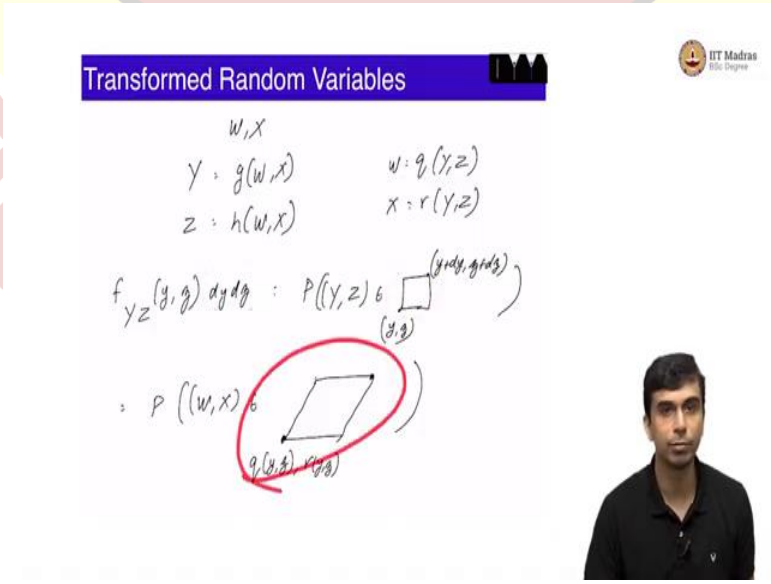
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Transformed Random Variables

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So, with that we have studied the covariance matrix and that finishes the study of covariance. We will move to the next and final topic in multiple random variables which is transformed random variables.

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Let us say you have two random variables W and X, they are general random variables, they can be independent, dependent whatever W and X are two random variables. Now, we define two other random variables Y and Z; Y is a function of w and x this is Y is g (w, x) and Z is h (w, x). You can ask the question, can you give the density of Y and Z in terms of the density of W and X?

The question is, what is the joint density of Y Z? Let us say you have the joint density of W and X can you give the joint density of Y Z, in general it is not possible, in general it might be quite complicated. But for a special case where this happens to be a invertible transform, what do I mean by invertible transform; by that I mean that from YZ you can get back W and X, from W,X you got Y and Z because that is the definition.

Y and Z are functions of W and X but if they happen to be invertible from Y and Z you can get back W and X. So, that means that there exists some functions Q and R such that W is equal to q of y, z and x is equal to r of y, z. So, this q and r are effectively inverses of g and h so g and h transform W X to Y Z, q and r transform Y Z to W X.

So, if this these q and r not always exist right you can clearly imagine a case where it is not possible, why is that? Let us say if g of w , x was w and H of W - were also W right so let us take the extreme case where g of w , x is W; h of W , X is W, in this case y is always going to be equal to Z which is also going to be always going to be equal to W.

That means that from W and X you can get back Y and Z but from y and z you cannot get back W and X because X is effectively lost, so y equal to z equal to w, so x you cannot recover x from y and z. So, in such a situation it is not an invertible transform but for cases where g and h are invertible you can get back that is you can get back w and x from y and z there is a pretty important rule which is called the change of variable rule in at least in probability.

It is related to the change of variable rule in calculus will do that I am not going to derive this entirely but I will give the broad intuition behind this. So, what is f YZ of Y, Z, this is equal to the probability of y ,z the vector y , z belonging to this particular rectangle.

The one corner is y, z, this corner is y + d y , z + t z and this is a semi graphical semi symbolic derivation I am not going to derive this, this is just going to illustrate what we have to do. So, f_Y is simply the probability of Y, Z, Y belonging to y + d y and z belonging to z + b z divided by dy dz.

So, expressing that in a graphical way this is equal to probability of y , z belonging to the rectangle given by this where one end is y , z, this other diagonal is y + d y , z + d z of course this would be y + d y , z and this would be y , z + d z that would be the 4 points of this rectangle.

So, what is the probability of y , z belongs to this? Here is where you can use the invertibility of y , z and w , x that is you happen to know the density of w , x that was the starting premise of this problem so can you get the density of y and y , z from the density of w , x? And because this happens to be an invertible thing so probability of y, z belonging to this particular square is exactly equal to probability of W, X.

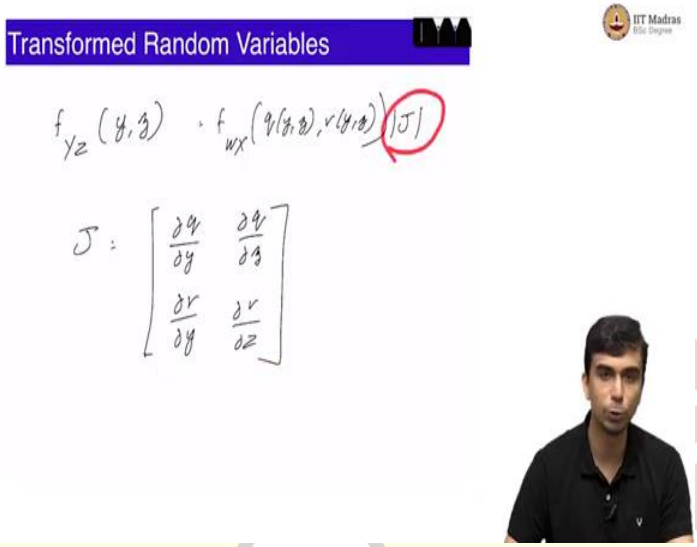
Let us move the dy dz to the left side so that we have some space, so the probability of y , z belonging to this particular rectangle is exactly equal to the probability of w , x belonging to will not call it a rectangle we will just do the 4 points. So, y , z should be transformed by q , r into as follows, it is going to be q of y , z into r of y , z, it is actually going to be a parallelogram am going to give the answer away already.

So, what is this, the left most point here is simply q of y , z , r of , z, so that is the left point here. The right point is simply q of y + d y , z + d z and r of y + d y , z + d z and similarly you have these four points and these four points form a parallelogram you can easily see that they do not form a rectangle always but they always form a parallelogram.

So, now that is the key part here this even though you start off with the probability of your y , z belonging to the rectangle that is equal to the probability of w , x belonging to this parallelogram and so this means that you have this probability let us say the parallelogram is small enough, then this is exactly equal to the product, the density of w , x at the point q of y , z and r of y , z times the area of this parallelogram.

So, this d y d z corresponding to the area of this rectangle but on the right hand side you would have, if you had the density of w, x you have to have the area of this parallelogram and I am not going to go into the calculus part of it and the area of a parallelogram is related to this related to what is called the determinant of the Jacobian given by of this matrix.

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I am not going to go into the great details here but this effectively says that $f_{YZ}(y,z)$ is equal to dY dZ is equal f_{WX} of effectively the inverse of y , z, so y , z is some numbers you can invert y , z to get w , x; w , x of its q of y , z; r of y , z but it does not stop here because here left hand side you had the area of the square, area of the rectangle but on the right hand side you have area of this parallelogram.

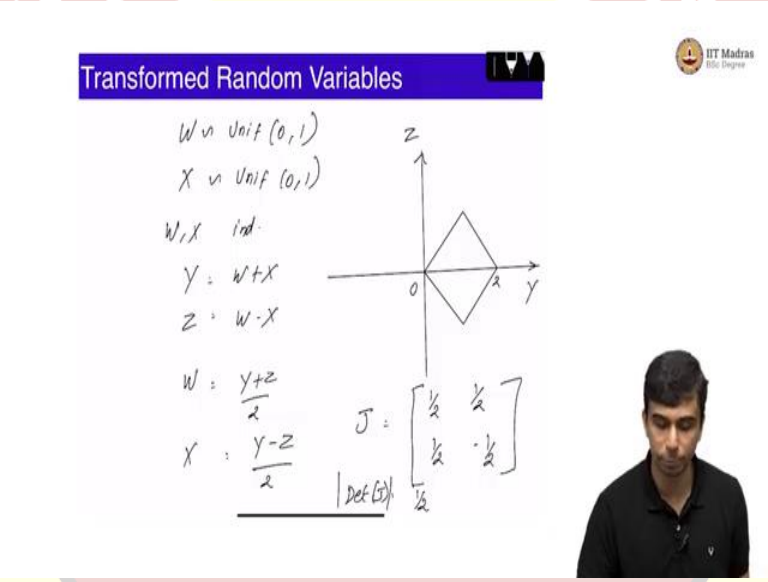
And that area of the parallelogram is determinant of the Jacobean, what is this Jacobean? J is the matrix given by as follows; you take the operator which converts Y, Z to W, X and you can write that as $\begin{bmatrix} \frac{\partial q}{\partial y} & \frac{\partial q}{\partial z} \\ \frac{\partial r}{\partial y} & \frac{\partial r}{\partial z} \end{bmatrix}$ this matrix is called the Jacobean which is effectively collecting the gradients of these two functions q and r are two functions and it collects the gradients of these two functions in its row of operators and it is in zero matrices and zero vectors is called the Jacobean matrix.

And the determinant of this Jacobean is intimately related to the area of this parallelogram, I am not going to great detail into showing this but why do you get this Jacobean; that is because of the you want to relate the area of this parallelogram to you want to get the area of this parallelogram and that is effectively given by the determinant of this Jacobean.

I hope you know what the determinant means, determinant of a at least for 2 ×2 matrices it is pretty simple, in this case it is $\frac{\partial q}{\partial y} \times \frac{\partial r}{\partial z} - \frac{\partial q}{\partial z} \times \frac{\partial r}{\partial y}$ this is called the change of variable rule in to get the density of derived random variables from original random variables.

So, this can be sorry into d by d z, so the d y d z can be cancelled on both sides which would effectively mean that you have the density of this random variable y and z the joint density of the random variables y and z is given by the joint density of random variables w and x evaluated at q of y , z and r of y , z times the Jacobean, this Jacobean can depend on y and z also right, because you can see that the Jacobean is the derivative of q with respect to y and z and r with respect to y and z, the Jacobean can also depend on y and z.

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Let us do a very simple couple of examples, let us take the simplest example that we can think of W, so let us say W is distributed as uniform (0 ,1) and z X is distributed uniform (0 ,1) and let us say they are also independent, are independent W X independent. Now, we will define a two pair of random variables Y and Z.

Say Y is equal to (W + X) and Z is equal to (W – X) , of course in this particular example you do not need to go to the extreme of the using the change of variable rule but change the variable rule gives a very elegant answer here and it is applicable for more complex cases also. So, you have two random variables y and z y is equal to (w + x) and z is equal to (w – x) you will see that even though the w and x are independent random variables y and z are not independent random variables you will see that pretty easily, why is that?

Well you can draw a pretty simple curve here, so y which is (w + x) can take values in the range of 0 to 2 and z which is w - x can take values in the range of (- 1 to 1). In fact, so (w + x) and (w

– x) the joint distribution will actually be non-zero in this diamond not being very exact the diamond shape here this is y and this is z.

So, outside this diamond y and z density of y z is going to be 0 and within this diameter it is going to be non-zero, why is that? Well that is follows simply from the logic of y being w + x and z being w - x and you can clearly see that y and z are not independent I mean a characteristic feature of an independent kind of random variables is that the support or the region where it is nonzero is a perfect rectangle or union of rectangles.

It cannot have diamond shape things where you have this region is has 0 joint density but it has marginals are not zero right both the marginals are not zero but the joint is zero here. So, this clearly is not a independent pair of random variables. So, now that is aside now, now what we want is the joint distribution or the joint density of y and z, how do we get that?

Well here is where we can use our change of variable rule; the first question you should ask is, is this an invertible transform? From w and x we got y and z but from y and z can we get back w and x? Yes, easily we can. W is equal to (y + z) / 2 and x is equal to (y – z) / 2. So, this is a pretty easy inverse to get you can from y and z you can get back w and x, why is that?

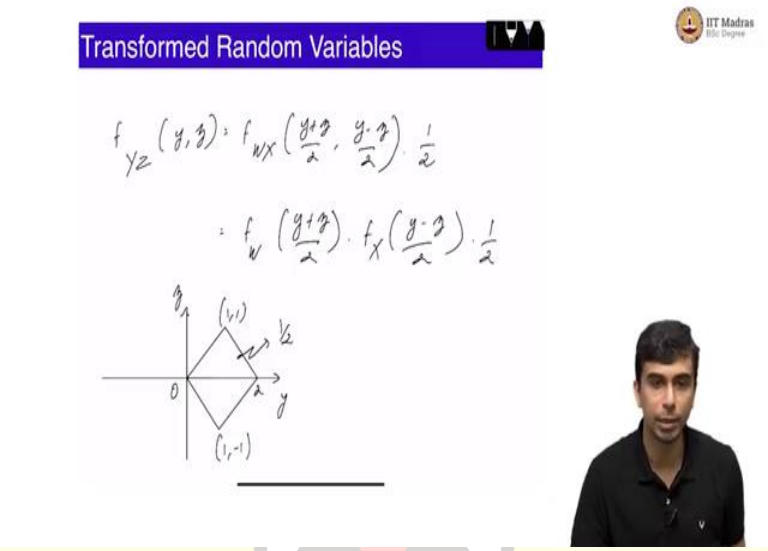
Because (y + z) / 2 would simply be w and (y – z)/ 2 would simply be x, this is a pretty easy. So, now we have that this q and r functions do exist so q of y , z is simply (y + z)/ 2 and r of y , z is simply (y - z) / 2 with that can we get the Jacobean now; what is the Jacobean; recall Jacobean has this dou q by dou y.

So, q is (y + z) / 2, so $\frac{\partial q}{\partial y}$ is simply 1 / 2 and similarly $\frac{\partial q}{\partial z}$ is also equal to 1/ 2, what about dou r; r of y , z it is (y – z) /2, $\frac{\partial r}{\partial y}$ is 1/2 and $\frac{\partial r}{\partial z}$ is – 1/2. So, because q is equal to (y + z) / 2 and r is equal to q of y , z is y + z by 2 and r of y , z is y - z by 2. So, we have the Jacobean as this and what is the absolute value of the determinant of this Jacobean?

We can compute that pretty easily which is going to be equal to ah determinant of Jacobean here is simply 1 by 4, - 1 by 4 which is equal to 1/ 2 are the absolute values 1/2. If you computed the values – 1/ 4 – 1/4 this is - 1 /2 but we are only interested in the absolute value because area cannot

be negative. So, the absolute value is simply equal to 1/ 2, so that is we have the determinant of the Jacobean as 1 / 2.

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Now, we can go back to get our density functions, we have $f_{Y\,Z}(y, z)$ is equal to $f_{W\,X}$ of you have to apply the inverse q and r to y and z which is going to be equal to (y + z) /2 , (y - z)/ 2 into the determinant of the Jacobean absolute value of the determinant of the Jacobean which is 1 by 2.

And f_{WX} of (())(16:55) - z / 2 is particularly nice because we assume that w and x are independent random variables and because w and x are independent random variables we have this f w of (y + z)/ 2 into f x of (y – z) / 2 × 1 / 2 and if you evaluate this when is (y + z) / 2 when is this term nonzero; because w and x were uniform 0 1 there are only two values f_W and f_X takes either 1 or 0.

So, you just need to ask the question when is fw of (y + z) / 2 non-zero? Well f w of (y + z) / 2 non-zero if (y + z) /2 is between 0 to 1, when is f x of (y – z) /2 non-zero? Well when (y – z) / 2 is between 0 and 1. If you put all of that together this entire f y z of (y , z) is non-zero when (y + z)/2 is between 0 and 1 and (y – z) / 2 is between 0 and 1. If you when is when is that going to happen?

So, this is the y axis, this is the z axis (y + z) / 2 is between 0 and 1 and (y – z) / 2 is also between 0 and 1. If you put that both together you get (2 ,0) (1 , 1) and (1 ,- 1). So, within this diamond it is nonzero and what is the value here; the value is going to be this is 1, this is 1, this is half.

So, within this diamond the value is going to take a value of half and outside this diamond it is going to take a value of 0. So, this is a pretty simple example but you can apply this to even pretty given complicated examples where you do not have constant Jacobeans but rather Jacobeans which are functions of y and z. The reason why we are doing this is because this will be a useful tool for studying multivariate normal distributions. With that we can wrap up this chapter where we study multiple random variables.

